Accuracy vs Precision

- In chemistry, the meanings of accuracy and precision are quite different.

- **Accuracy** is a measure of how close a measurement comes to the actual or true value of whatever is measured.

- **Precision** is a measure of how close a series of measurements are to one another, irrespective of the actual value.

Accuracy vs Precision

- To evaluate the **accuracy** of a measurement, the measured value must be compared to the correct value. Or,
- **Accuracy** is related to the closeness to the correct/true value.

- To evaluate the **precision** of a measurement, you must compare the values of two or more repeated measurements. Or,
- **Precision** is related to the closeness to other estimated/measured values.

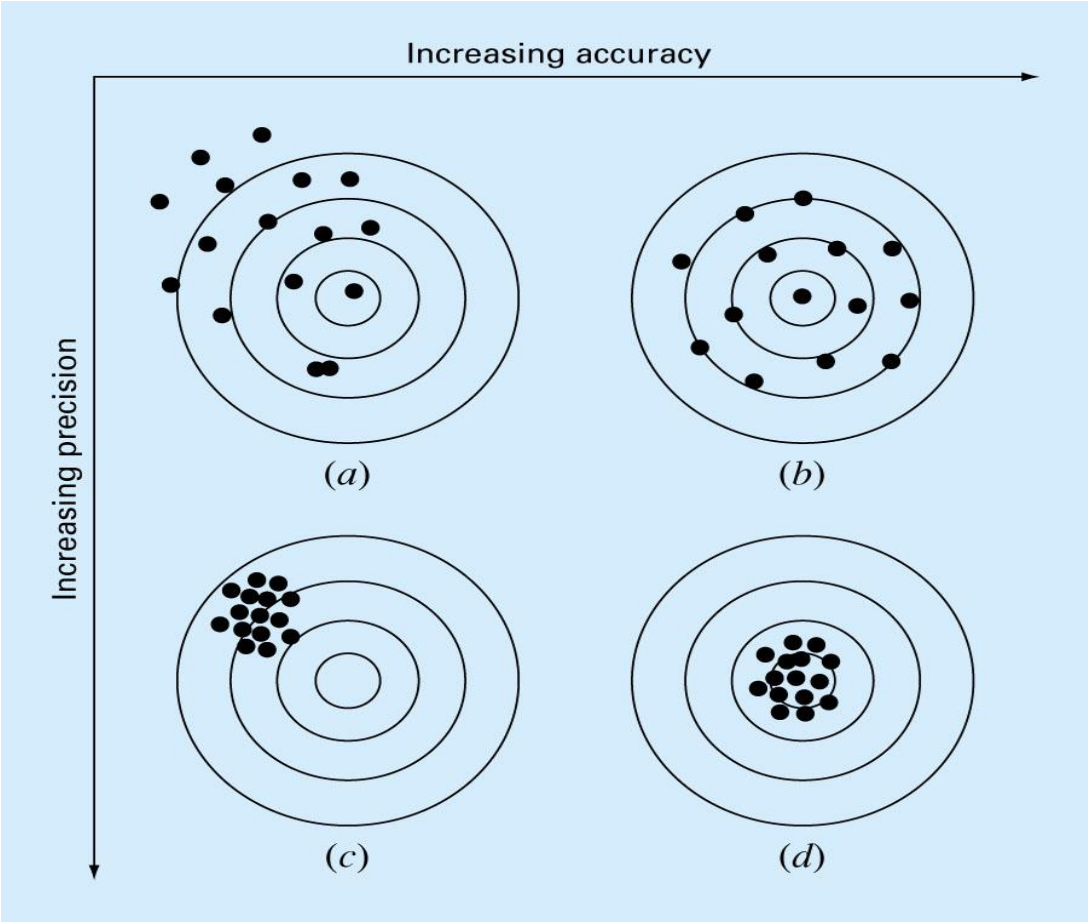
Accuracy vs Precision

Darts on a dartboard illustrate the difference between accuracy and precision.

Good Accuracy, Good Precision | **Poor Accuracy, Good Precision** | **Poor Accuracy, Poor Precision**

The closeness of a dart to the bull's-eye corresponds to the degree of accuracy. The closeness of several darts to one another corresponds to the degree of precision.

Accuracy vs Precision



An example from marksmanship illustrating the concepts of accuracy and precision: (a) inaccurate and imprecise, (b) accurate and imprecise, (c) inaccurate and precise, and (d) accurate and precise.

Error

Determining Error

- There is a difference between the **accepted value**, which is the correct value for the measurement based on reliable references, and the **experimental value**, the value measured in the lab.

- The difference between the experimental value and the accepted value is called the **error**.

Error

Definitions

- Numerical errors arise from the use of approximations to represent exact mathematical operations and quantities.
- For such errors, the relationship between the exact, or true, result and the approximation can be formulated as
- True value – Approximation = Error
- $\mathcal{E}_t$ = Accepted value – Experimental value
- $\mathcal{E}_t$ is used to designate the exact value of the error

Error

Determining Error

Note that the true error is commonly expressed as an absolute value and referred to as the absolute error

$$\text{Absolute error} = |\text{accepted} - \text{measured}|$$

$$\text{Relative error} = \frac{|\text{measured} - \text{accepted}|}{\text{accepted value}}$$

$$\text{Percent relative error } (\varepsilon_a) = \frac{|\text{measured} - \text{accepted}|}{\text{accepted value}} \times 100\%$$

Error

Determining Error

- Suppose you use a thermometer to measure the boiling point of pure water at standard pressure.
- The thermometer reads 99.1°C.
- You probably know that the true or accepted value of the boiling point of pure water at these conditions is actually 100.0°C.

Error

Determining Error

- For the boiling-point measurement, the error is 99.1°C – 100°C, or –0.9°C.

- For the boiling-point measurement, the error is 100°C – 99.1°C, or 0.9°C.

Error

Calculating Percent Error

The boiling point of pure water is measured to be 99.1^0 C. Calculate the percent error.

Error

Solve for the unknown.

Substitute the equation for error, and then plug in the known values.

$$\text{Percent error } (\varepsilon_a) = \frac{|\text{experimental value} - \text{accepted value}|}{\text{accepted value}} \times 100\%$$

$$= \frac{|99.1\,^\circ C - 100.0\,^\circ C|}{100.0\,^\circ C} \times 100\%$$

$$= \frac{0.9\,^\circ \cancel{C}}{100.0\,^\circ \cancel{C}} \times 100\ \% = 0.9\%$$

Error

- Often, when performing computations, we may not be concerned with the sign of the error but are interested in whether the absolute value of the percent relative error is lower than a prespecified tolerance ε_s. Therefore, it is often useful to employ the absolute value for expressing
- For such cases, the computation is repeated until

$$\varepsilon_a < \varepsilon_s$$

- This relationship is referred to as a stopping criterion.
- If it is satisfied, our result is assumed to be within the prespecified acceptable level

Error

- For numerical methods, the true value will only be known when we deal with functions that can be solved analytically.
- However, in real-world applications, we will obviously not know the true answer a priori.
- For these situations, an alternative is to normalize the error using the best available estimate of the true value, that is, to the approximation itself, as

$$\varepsilon_a = \frac{\text{Approximate error}}{\text{Approximation}} \text{ X } 100\%$$

Error

- One of the challenges of numerical methods is to determine error estimates in the absence of knowledge regarding the true value.
- For example, certain numerical methods use iteration to compute answers.
- In such cases, a present approximation is made on the basis of a previous approximation.
- This process is performed repeatedly, or iteratively, to successively compute (hopefully) better and better approximations.

Error

- For such cases, the error is often estimated as the difference between the previous and present approximations.
- Thus, percent relative error is determined according to;

$$\varepsilon_a = \frac{\text{Present Approximation} - \text{Previous Approximation}}{\text{Present Approximation}} \times 100\%$$

- Often, when performing computations, we may not be concerned with the sign of the error but are interested in whether the absolute value of the percent relative error is lower than a prespecified tolerance $\boldsymbol{\varepsilon}_s$
- For such cases, the computation is repeated until

$$|\varepsilon_a| < \varepsilon_s$$

Error Estimation

- In mathematics, functions can often be represented by infinite series. For example, the exponential function can be computed using Maclaurin series expansion

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots + \frac{x^n}{n!} \quad ..(1)$$

- Thus, as more terms are added in sequence, the approximation becomes a better and better estimate of the true value of e^x

Error Estimation

- It can be shown (Scarborough, 1966) that if the following criterion is met, we can be assured that the result is correct to at least *n* significant figures.

$$\varepsilon_s = (0.5 \times 10^{2-n})\%$$

- This can be employed to determine the error criterion that ensures a result that is correct to at least three significant figures

$$\varepsilon_s = (0.5 \times 10^{2-3})\% = 0.05\%$$

Error Estimation

- Starting with the simplest version, $e^x = 1$, add terms one at a time in order to estimate $e^{0.5}$.
- The true value for $x=0.5$ is $e^{0.5} = 1.648721 \ldots$
- *For* $n=1$, $e^{0.5} = 1 + 0.5 = 1.5$
- This represents a true percent relative error

$$\varepsilon_t = \frac{1.648721 - 1.5}{1.648721} \text{ X } 100\%$$

$$\varepsilon_t = 9.02\%$$

$$\varepsilon_a = \frac{\text{Present Approximation} - \text{Previous Approximation}}{\text{Present Approximation}} \text{ X } 100\%$$

$$\varepsilon_a = ((1.5 - 1.0)/1.5)x100\% = 33.3\%$$

Error Estimation

Terms	Result	$\varepsilon t\%$	$\varepsilon a\%$
1. n=0	1	39.3	
2. 1	1.5	9.02	33.3
3. 2	1.625	1.44	7.69
4. 3	1.645833333	0.175	1.27
5. 4	1.648437500	0.0172	0.158
6. 5	1.648697917	0.00142	0.0158

True value 1.648721

Thus, after six terms are included, the approximate error falls below ε_s = 0.05%, and the computation is terminated.

Computer Application

- To do this for the same problem as above example, the series expansion can be expressed as

$$e^x = \sum_{i=0}^{n} \frac{x^i}{i!}$$

```
• #include <stdio.h>
• #include <stdlib.h>
• #include <math.h>
• double fact(double n){ //Calculating
  factorial
•     double result = 1.0;
•     for(double i=1;i<=n;i++){
•         result *= i;
•     }
•     return result;
• }
```

```
• int main(){
•     double np,n,et,es,ea,tv = 1.648721,pv,pri_v,x=0.5; //e power 0.5
•     printf("Enter the degree of precision : ");
•     scanf("%lf",&np);
•     es = (0.5 * pow(10,(2-np)));
•     printf("Precision %.10f and Stopping criteria = %.10f\n",np,es);
•     printf("True Result = %g\n\n",tv);
•     int i=0; ea=1; pri_v=0; //initial estimation and previous value
•     while(1){
•         pv += (pow(x,i)/fact(i));
•         et = (fabs(tv - pv)/tv)*100; //using true value
•         ea = (fabs(pv - pri_v)/pv)*100;  //using present approximation
•         printf("n=%d,\tResult = %.10f,\tet = %.10f,\tea = %.10f\n",
  i,pv,et,ea);
•         if(et<es) break;
•         i++; pri_v = pv;
•     }
•     return 0;
• }
```

6 digits

Select E:\Daffodils\Spring_2022\Error_0\bin\Debug\Error_0.exe

```
Enter the degree of precision : 3
Precision 3.000000 and Stopping criteria = 0.050000
True Result = 1.64872

n=0,    Result = 1.000000,      et = 39.346924, ea = 100.000000
n=1,    Result = 1.500000,      et = 9.020386,  ea = 33.333333
n=2,    Result = 1.625000,      et = 1.438752,  ea = 7.692308
n=3,    Result = 1.645833,      et = 0.175146,  ea = 1.265823
n=4,    Result = 1.648438,      et = 0.017195,  ea = 0.157978

Process returned 0 (0x0)   execution time : 1.896 s
Press any key to continue.
```

10 digits

E:\Daffodils\Spring_2022\Error_0\bin\Debug\Error_0.exe

```
Enter the degree of precision : 6
Precision 6.0000000000 and Stopping criteria = 0.0000500000
True Result = 1.64872

n=0,    Result = 1.0000000000,  et = 39.3469240702,     ea = 100.0000000000
n=1,    Result = 1.5000000000,  et = 9.0203861054,      ea = 33.3333333333
n=2,    Result = 1.6250000000,  et = 1.4387516141,      ea = 7.6923076923
n=3,    Result = 1.6458333333,  et = 0.1751458656,      ea = 1.2658227848
n=4,    Result = 1.6484375000,  et = 0.0171951470,      ea = 0.1579778831
n=5,    Result = 1.6486979167,  et = 0.0014000752,      ea = 0.0157952930
n=6,    Result = 1.6487196181,  et = 0.0000838192,      ea = 0.0013162571
n=7,    Result = 1.6487211682,  et = 0.0000101991,      ea = 0.0000940183

Process returned 0 (0x0)   execution time : 4.803 s
Press any key to continue.
```

Learning Outcomes

- Familiar with error
- Computer implementation (a problem)